

READING PASSAGE 3

The Physics of Tsunamis

A The study of the physics of tsunamis has undergone a remarkable transformation over the past several decades. What was once a niche area of academic inquiry has developed into a rich and diverse field of research, drawing on insights from a wide range of disciplines and theoretical traditions.

B The origins of contemporary research in this area can be traced to pioneering studies conducted during the mid-twentieth century. These early investigations, though constrained by the technological and methodological limitations of their era, established foundational principles that continue to inform current scholarship. Successive generations of researchers have built upon these foundations, progressively refining theoretical models and expanding the empirical evidence base through increasingly sophisticated research designs.

C At the core of current understanding lies a set of interrelated mechanisms that collectively explain the observed patterns and outcomes. First, structural factors create the underlying conditions within which individual and collective action takes place. These structural dimensions include economic incentives, institutional frameworks, and cultural norms that shape behaviour in ways that are often not immediately apparent to the actors themselves. Second, dynamic feedback processes amplify or dampen initial effects, generating patterns of change that can diverge substantially from what linear models would predict.

D Case studies from diverse geographical and cultural settings provide compelling illustrations of these dynamics in action. In one particularly well-documented example, researchers tracked developments over a fifteen-year period in a setting that offered a natural experiment in the effects of policy change. The introduction of a new regulatory framework produced a cascade of adaptations that rippled through the entire system, generating outcomes that both confirmed theoretical predictions and revealed previously unrecognised complexities.

E Several unresolved debates continue to animate scholarly discussion in this field. Perhaps the most prominent concerns the relative importance of structural versus agentic factors in determining outcomes. While some researchers emphasise the constraining power of institutional and economic structures, others highlight the capacity of individuals and groups to exercise agency and effect change even within seemingly restrictive contexts. This debate has significant practical implications, as it shapes recommendations for policy interventions and institutional reforms.

F The implications of this body of research extend well beyond the academy. As societies grapple with increasingly complex challenges, the insights generated by scholarly inquiry in this domain have the potential to inform more effective and equitable policy responses. Realising this potential, however, will require sustained investment in both fundamental research and the translational infrastructure needed to convert knowledge into action.

G In conclusion, the study of the physics of tsunamis represents a vital and dynamic area of contemporary research. The progress that has been made in understanding the key mechanisms and dynamics provides a solid foundation for both further scholarly inquiry and practical application. The challenges that remain should be seen not as reasons for discouragement but as opportunities for continued intellectual growth and social contribution.

Question 1

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 1 Contemporary understanding of this topic is more sophisticated than it was fifty years ago.

Question 2

Choose the correct letter, A, B, C or D.

- 2 The author suggests that early studies in this field were limited by:
- A. excessive research funding
 - B. translating academic findings into practical policy
 - C. lack of public interest
 - D. too many international collaborations

Question 3

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 3 Dynamic _____ processes can amplify or dampen initial effects in complex systems.

Questions 4–5

Choose the correct letter, A, B, C or D.

- 4 Which of the following best describes the author's view on future developments?
- A. economic incentives and institutional constraints
 - B. individual personality traits
 - C. random chance
 - D. government policies alone
- 5 The author suggests that early studies in this field were limited by:
- A. excessive funding
 - B. theoretical and methodological limitations
 - C. public interest
 - D. international collaboration

Question 6

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 6 The passage suggests that structural factors are less important than individual choices.

Questions 7–9

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 7 The passage identifies power _____ as a key factor at the organisational level.
- 8 According to the text, cognitive processes shape the _____ that people make.
- 9 According to the passage, structural factors include economic incentives, institutional frameworks, and _____.

Question 10

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 10 The author believes that all research questions in this field have been answered.

Question 11

Choose the correct letter, A, B, C or D.

- 11 What finding emerged from the landmark study discussed in the passage?
- A. excessive funding
 - B. theoretical and methodological limitations
 - C. public interest
 - D. international collaboration

Question 12

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 12 The landmark study discussed in the passage involved participants from multiple countries.

Question 13

Choose the correct letter, A, B, C or D.

- 13 The meta-analyses cited in the passage reveal that:
- A. pessimism about the future
 - B. cautious optimism with acknowledgment of challenges
 - C. complete certainty about outcomes
 - D. indifference to practical applications